第 1 章 Maximum Likelihood Estimation

1.1 Likelihood function and the ML principle

Let $f(y;\theta_0)$ denote the probability density function of the random variable y given θ_0. Our objective is to estimate the true parameter vector θ_0. The joint density of n iid observations of y is

$$f(y_1,\ldots,y_n|\theta_0)=\prod_{i=1}^{n} f(y_i;\theta_0) \tag{1}$$

Now taking $f(y_i;\theta)$ as a function of θ given $y_1,\ldots,y_n$ and write

$$L(\theta|y_1,\ldots,y_n)=\prod_{i=1}^{n} f(y_i;\theta) \tag{2}$$

gives the **likelihood function**, the likelihood that the population parameter is θ given the observed sample.

1.1.1 ML principle

The **Maximum Likelihood (ML) principle** suggests that estimators of the unknwon parameters are obtained by **maximizing $L(\theta)$ with respect to θ**. If the random variable y is discrete, $\hat{\theta}$ yields the value of θ that maximizes the probability of observing the sample that actually occurred.

Usually, $\hat{\theta}$ can be obtained by solving the likelihood equation

$$\frac{\partial \log L(\theta)}{\partial\theta}|_{\hat{\theta}}=0 \tag{3}$$

where $\log L(\theta|y_1,\ldots,y_n)=\sum_{i=1}^{n}\log f(y_i;\theta)$. Occationally the ML estimator is not unique and $\log L(\theta)$ may have only one global maximum, but multiple local maxima.

The **regularity conditions** are the following:

1. The first three derivatives of $\log f(y|\theta)$ with respect to θ are continuous and finite for almost all y and for all θ.
2. For all values of θ, $|\frac{\partial^3 \log f(y|\theta)}{\partial\theta_j\partial\theta_k\partial\theta_l}|$ is limited by a function that has finite expectations.
3. The domain of y does not depend on θ. For example of the uniform distribution, since the density function depends on θ_0, the theorem does not hold.
4. θ is an interior point to the compact parameter space Θ.

1.1.2 Bartlett Identities

Define the **score vector** $S(\theta)$ and the **Hessian matrix** $H(\theta)$ as

$$S(\theta) = \frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^{n} \frac{\partial \log f(y_i|\theta)}{\partial \theta} = \sum_{i=1}^{n} s_i(\theta) \tag{4}$$

$$H(\theta) = \frac{\partial^2 \log L(\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^{n} \frac{\partial^2 \log f(y_i|\theta)}{\partial \theta \partial \theta'} = \sum_{i=1}^{n} H_i(\theta) \tag{5}$$

- The **First Bartlett identity** states that $E[s_i(\theta_0)] = 0$. Hence, $E[S(\theta_0)] = 0$.
- The **Second Bartlett identity** states that $Var[s_i(\theta_0)] = -E[H_i(\theta_0)]$.

  证明.

  $$Var\left[\frac{\partial \log f(y_i|\theta)}{\partial \theta}\right] = -E\left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right]$$

  where y_i is a random variable. □

$Var[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)']$ defines the **Fisher's information matrix** denoted as $\mathcal{I}(\theta_0)$. Hence, the result $Var[s_i(\theta)] = E[s_i(\theta_0)s_i(\theta_0)'] = -E[H_i(\theta_0)]$ is also called the **information matrix identity**.

证明.

$$\begin{aligned} Var[s_i(\theta)] &= E[[s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0\text{, First Bartlett}}] \times [s_i(\theta_0) - \underbrace{E[s_i(\theta_0)]}_{=0\text{, First Bartlett}}]'] \\ &= E[s_i(\theta_0)s_i(\theta)'] \end{aligned}$$

□

1.2 Properties of MLE

Under the assumed regularity conditions of MLE possesses the following properties:

1. **Consistency** meaning $plim\hat{\theta} = \theta_0$
2. **Asymptotic normality** meaning $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1})$. The goal is to estimate the $[\mathcal{I}(\theta_0)]^{-1}$ matrix (Fisher's information matrix) $\to$ need to go to a known distribution (normal).
3. **Asymptotic efficiency** meaning that if $\tilde{\theta}$ is a regular consistent asymptotically normal estimator such that $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega)$, then $\Omega - [\mathcal{I}(\theta_0)]^{-1}$ is positive semi-definite, i.e., under these RC, the MLE asymptotically achieves the **Cramer-Rao** lower bound which is given by $[\mathcal{I}(\theta_0)]^{-1}$. Information matrix is smaller than Ω in the matrix sense, and the information matrix is the lower bound.
4. **Invariance** meaning that if $c(\theta)$ is a continuous and continuously differentiable one-to-one function, the MLE of $\gamma = c(\theta_0)$ is $c(\hat{\theta})$.

Now, don't forget that θ_0 is not known, so we want to estiamte the information matrix.

1.2.1 Estimators of the information matrix

There are three commonly used estimators of $\mathcal{I}(\theta_0)$:

- **Expected Information**: If the form of the expected values of the second derivatives of the log-likelihood function is known, then we can evaluate $\mathcal{I}(\theta)$ at $\hat{\theta}$. This rarely happens in practice.

- **Observed information**: Simply use $-\hat{H}(\hat{\theta})$.

  证明.

$$
\begin{aligned}
\mathcal{I}(\theta_0) = Var[s_i(\theta_0)] &= -E[H(\theta_0)] \\
&= Var\left[\frac{\partial \log f(y_i|\theta_0)}{\partial \theta_0}\right] \\
&= -E\left[\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right] \\
&\underset{\text{avg.}}{=} \frac{1}{N}\sum_{i=1}^{n}\left(\frac{\partial^2 \log f(y_i|\theta_0)}{\partial \theta \partial \theta'}\right) \\
&= -\hat{H}(\hat{\theta})
\end{aligned}
$$

  □

- **Outer Product of the Gradient (OPG)**: Because of the information matrix identity, we can also use $n^{-1}\sum_{i=1}^{n} s_i(\hat{\theta})s_i(\hat{\theta})'$.

  证明.

$$
\begin{aligned}
Var[s_i(\theta_0)] &= E[s_i(\theta_0)s_i(\theta)'] \\
&= \frac{1}{N}\sum_{i=1}^{N} s_i(\theta_0)s_i(\theta_0)' \\
&= \frac{1}{N}\sum_{i=1}^{N} s_i(\hat{\theta})s_i(\hat{\theta})'
\end{aligned}
$$

  □

  The OPG is notorious for its poor performance with finite samples.

1.3 Regressors